



ASME

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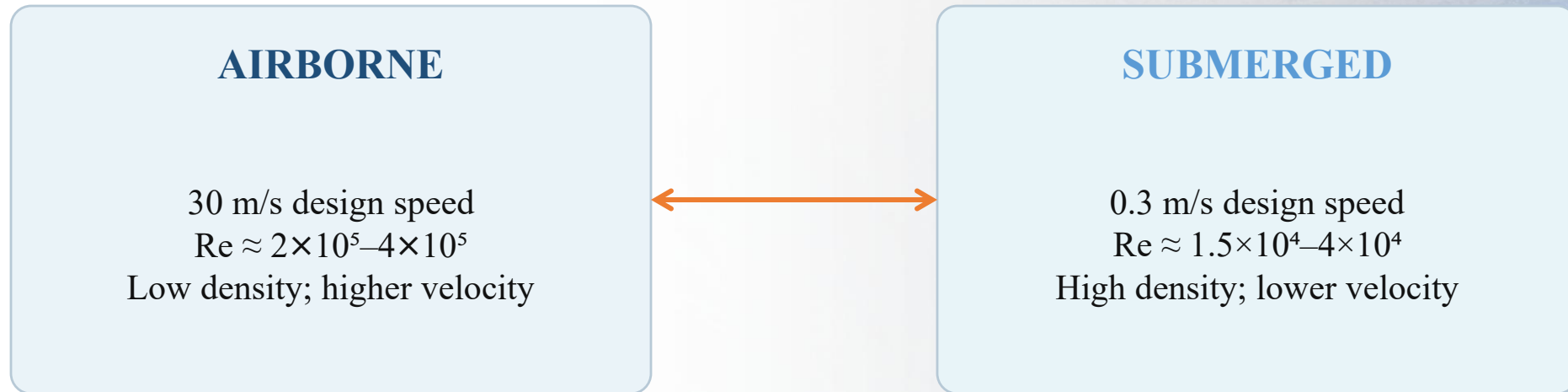


ONE WING FOR TWO WORLDS:
A BIO-INSPIRED DESIGN FOR CROSS-DOMAIN VEHICLES
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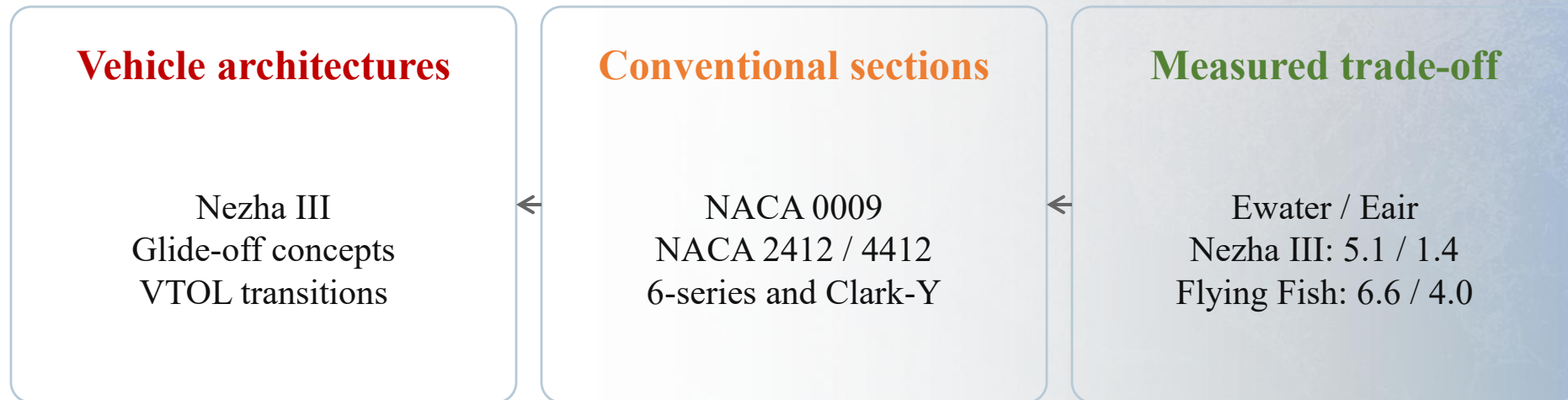
One Vehicle, Two Fluid Regimes

- A cross-domain vehicle must generate useful lift in air and water.
- Can we do that efficiently with one mechanically simplistic wing?



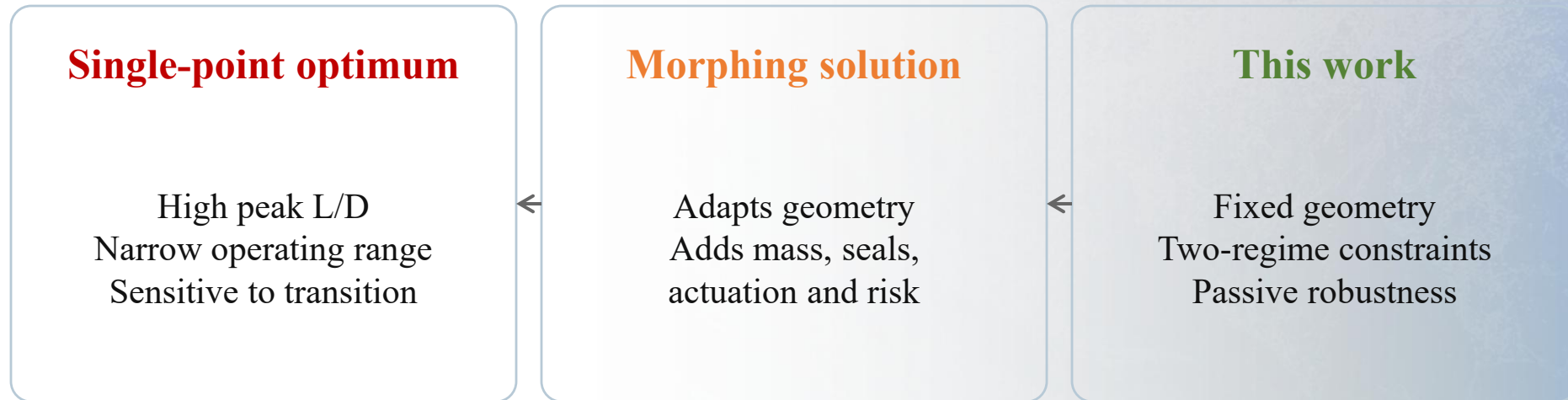
Prior Art Leaves the Airfoil as the Missing Design Variable

- Existing cross-domain vehicles vary their transition architecture, but most retain conventional airfoils and accept uneven efficiency.



The Design Gap: Efficiency Without Morphing

- The practical objective is not peak performance at one Reynolds number; it is acceptable efficiency across both operating envelopes.



Vehicle Requirements Define the Search Space

The airfoil is optimized within the geometry and operating envelope of the proposed cross-domain vehicle.

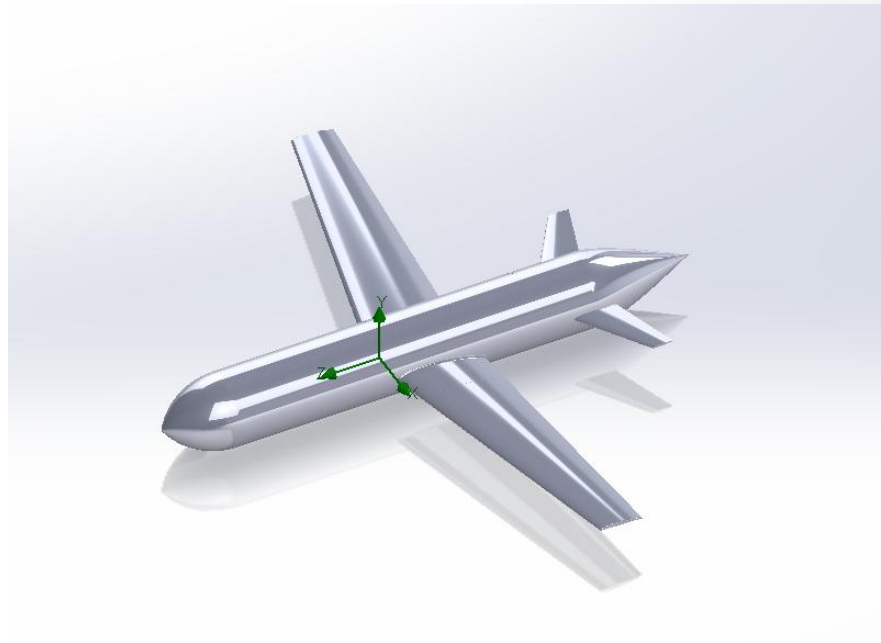


Figure 1. Cross-domain vehicle layout.

6.7

aspect ratio

156 mm

mean aerodynamic chord

1080 mm

wing span



Bio-Inspired Baseline: Puffin Wing Geometry

A puffin-inspired section supplies a biologically demonstrated starting point for flight and underwater propulsion.

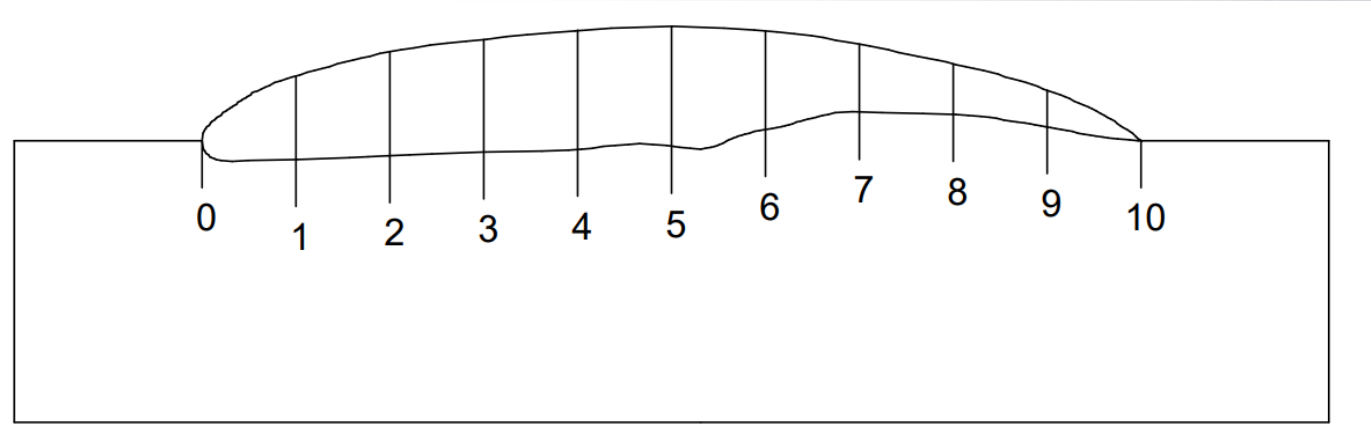


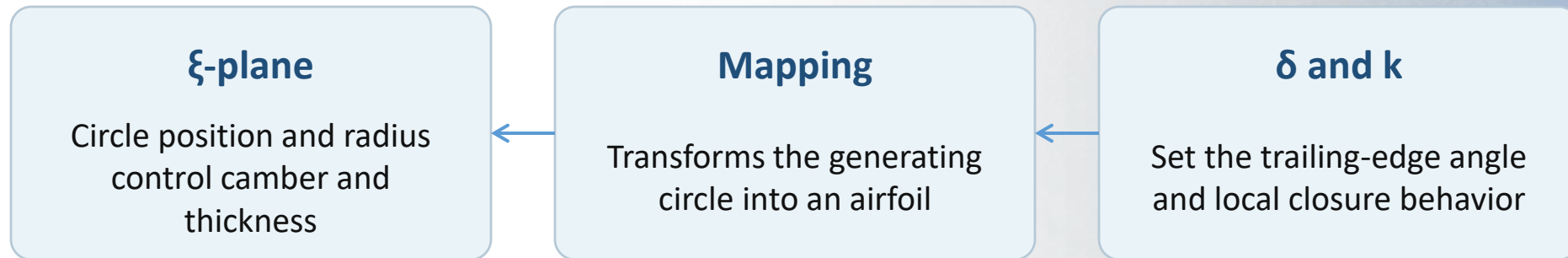
Figure 2. Bio-inspired baseline and controllable geometry.



Composite Karman–Trefftz Parameterization

- The mapping turns a compact parameter vector into a smooth, manufacturable airfoil family with explicit trailing-edge control.

$$z = kb \frac{(\xi + b)^k + (\xi - b)^k}{(\xi + b)^k - (\xi - b)^k}, \quad k = 2 - \frac{\delta}{\pi}$$



Alternating Optimization Across Two Reynolds Numbers

- Gradient-based updates alternate between representative low- and high-Reynolds-number conditions until both efficiency targets remain satisfied.

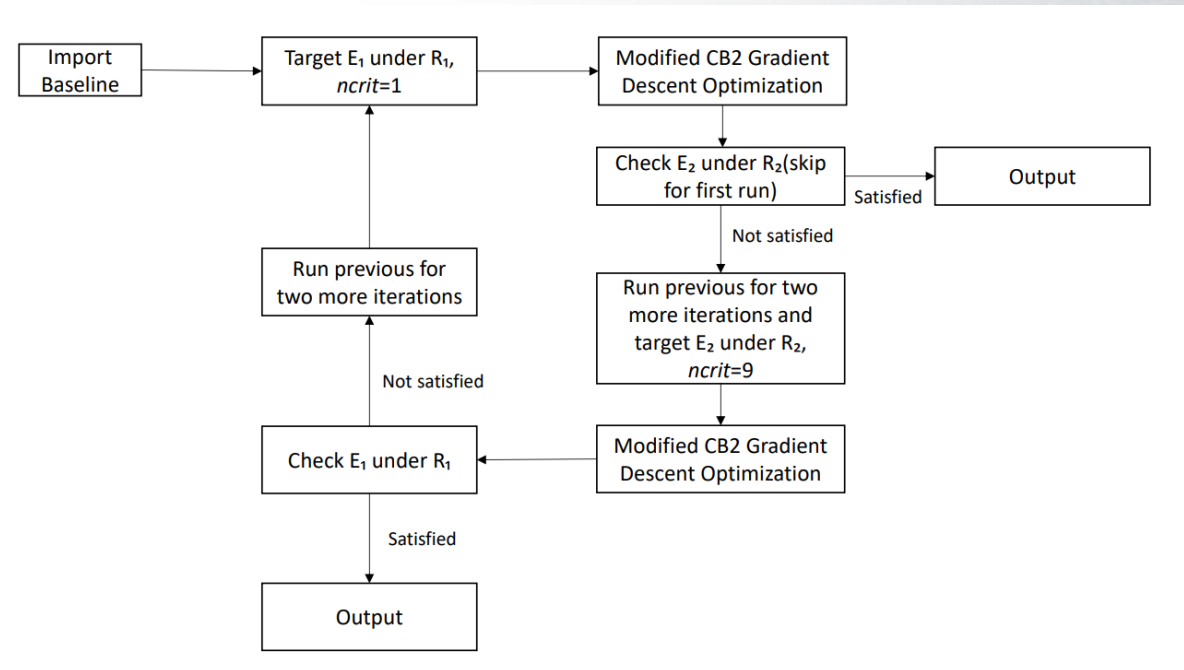


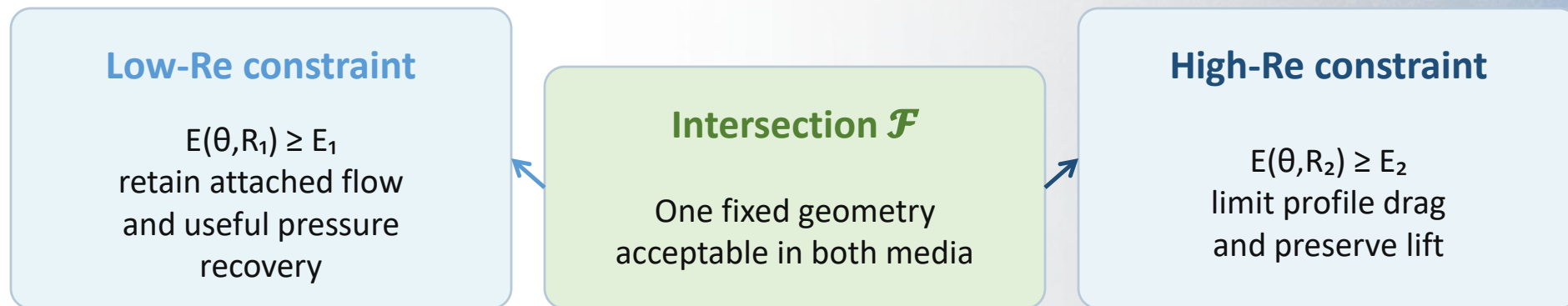
Figure 3. Five alternating cycles enforce two-regime performance.



Feasible Intersection

The alternating algorithm can be interpreted as searching for a geometry inside the intersection of two performance-feasible sets.

$$\mathcal{F} = \{\theta : E(\theta, R_1) \geq E_1\} \cap \{\theta : E(\theta, R_2) \geq E_2\}, \quad \theta^* \in \mathcal{F}$$

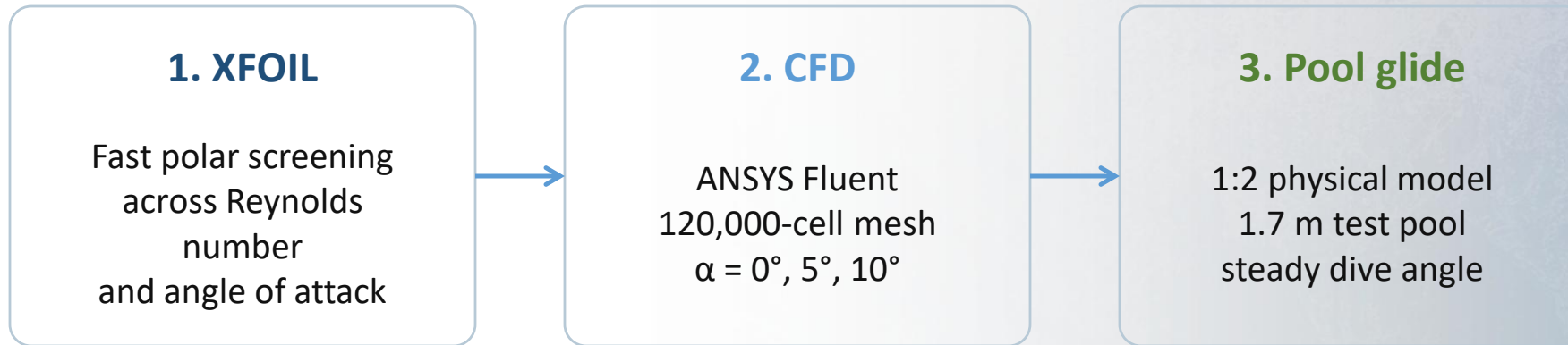


Deduction: robustness comes from constraint satisfaction, not from averaging the two fluids into one artificial operating point.



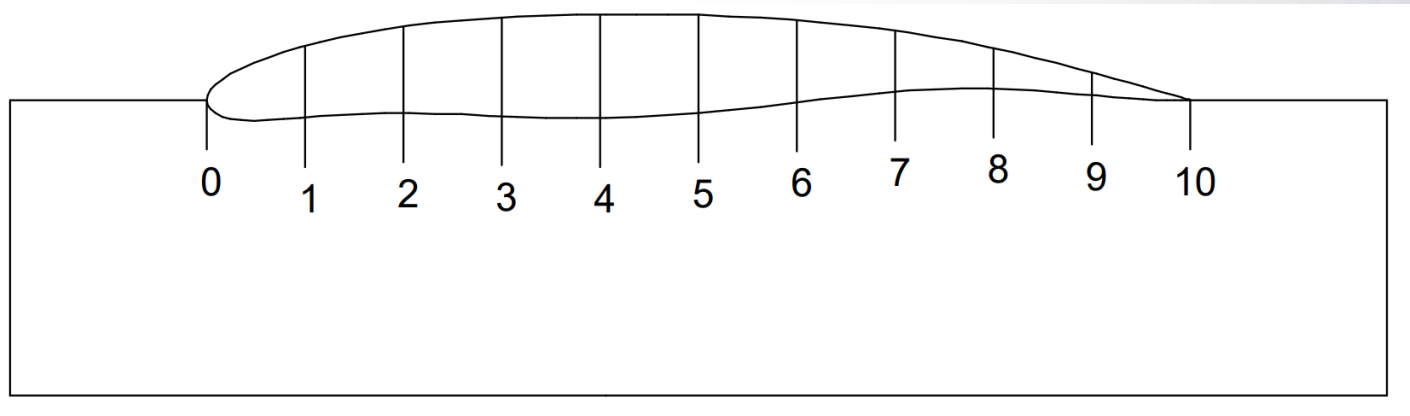
Validation Strategy

The design is screened rapidly, checked with higher-fidelity flow physics, then tested as a physical glider.



Optimized Geometry and Its Passive Mechanisms

The resulting section combines supercritical-like loading with a pronounced lower leading-edge cavity.



10.53%

maximum thickness
at 40.6% chord

4.00%

maximum camber
at 65.5% chord

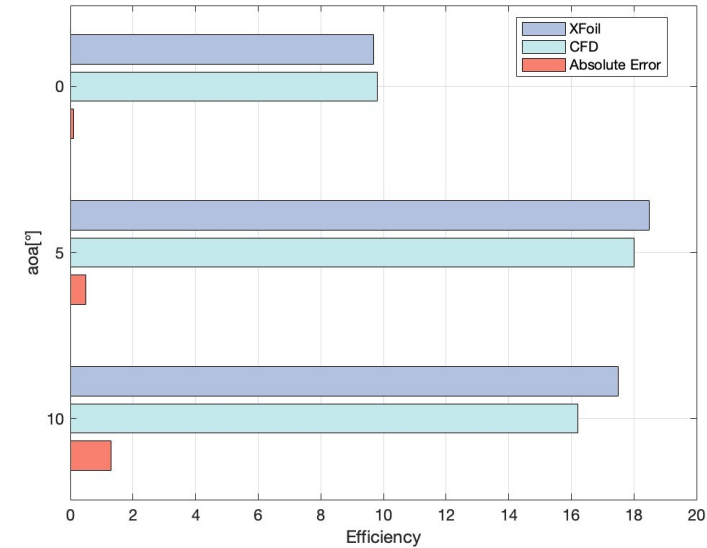
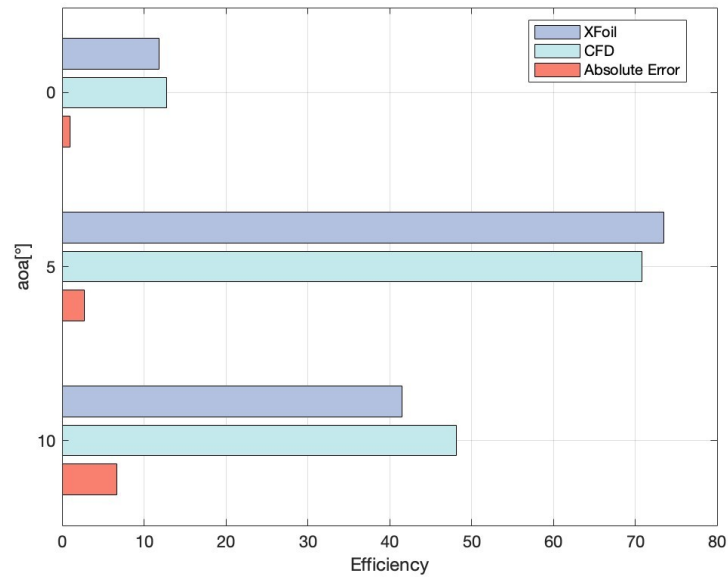
Figure 4. Optimized fixed airfoil.

Mechanistic reading:

- flatter upper mid-chord delays separation
- lower-surface concavity strengthens pressure difference
- local cavity effects are most valuable at low Re



CFD vs. XFOIL



Airfoil Results

- The optimized puffin section is compared with the best-performing symmetric and cambered NACA references at each Reynolds number.

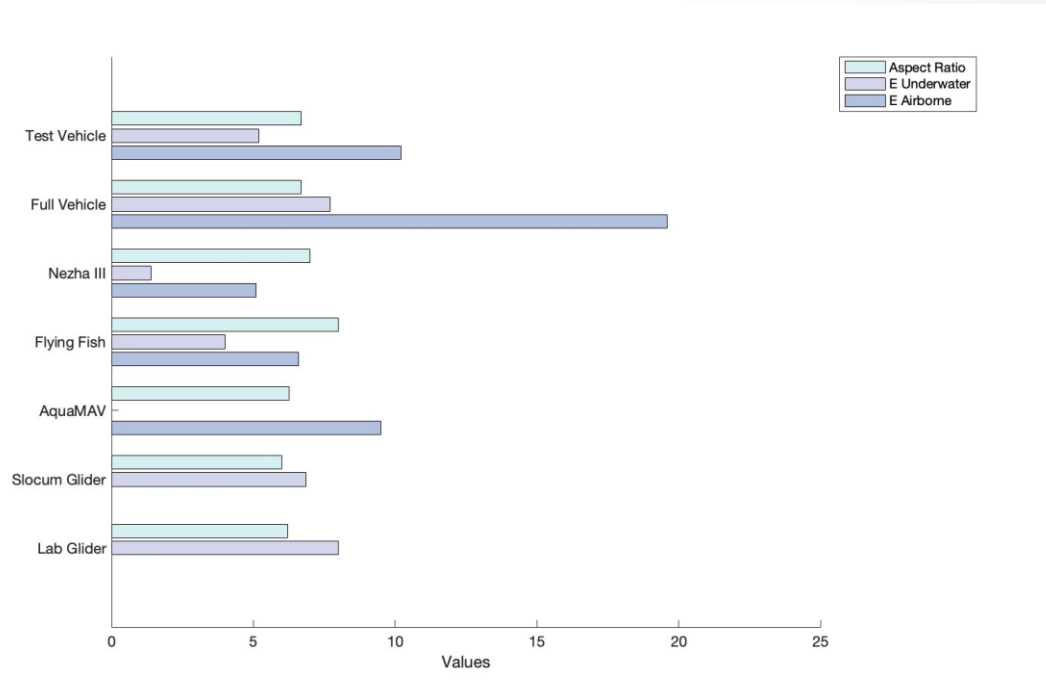
Re	Puffin E	Best symmetric E	Sym. gain	Best cambered E	Cambered gain
2×10^4	19.4	16.4	+18.3%	16.9	+14.8%
5×10^4	37.0	25.8	+43.4%	34.0	+8.8%
1×10^5	62.0	36.6	+69.4%	56.5	+9.7%
2×10^5	82.7	47.4	+74.5%	81.5	+1.5%

Within $2 \times 10^4 \leq Re \leq 2 \times 10^5$, the bio-inspired section consistently exceeds both comparison families; the advantage narrows as Re increases.



Vehicle-Level Comparison=

- The proposed vehicle is competitive underwater while retaining substantially higher airborne glide efficiency than prior cross-domain concepts.



Experimentation

- At static equilibrium, the measured dive angle gives the underwater glide efficiency directly.

$$C_L \sin \gamma = C_D \cos \gamma, \quad E \equiv \frac{C_L}{C_D} = \frac{1}{\tan \gamma}$$

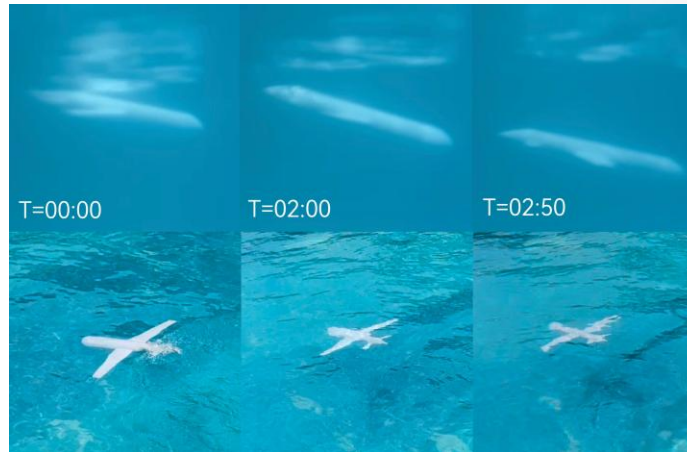


Figure 10. Transition to steady underwater glide.

10.1°

measured stable
dive angle

5.6

experimental
glide ratio

5.17–5.20

CFD / XFOIL
prediction



Limitations and Evidence Boundaries

- No wind-tunnel or full airborne polar validation is reported.
- Performance weakens beyond the intended Reynolds-number range; at $Re = 5 \times 10^5$, NACA 6412 is about 4.2% better.
- Transition, surface roughness, aeroelasticity and cross-domain impact loads require further study.



Conclusions and Design Implications

- A composite Karman–Trefftz representation turns bio-inspired geometry into an optimizable engineering design.
- Alternating two-Reynolds-number constraints locate a fixed geometry acceptable in both air and water.
- The optimized section leads conventional NACA references throughout the intended 2×10^4 – 2×10^5 range.
- CFD and XFOIL are most consistent at lift-producing angles; the zero-degree discrepancy remains a caution.
- Full vehicle shows competitive performance compared to morphing and conventional counterparts.
- Design implication: passive geometric robustness can substitute for part of the complexity of active morphing.





Thank you!

Questions?